Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.

Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.

This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

3.3 Observed Results: Dataset Split and Sample Images

GitHub Repository: https://github.com/Hasini923/Deep_Learning/tree/main/Attention

The `flower_photos` archive of the TensorFlow Flowers dataset (3,670 images, the same images as `tf_flowers`) was used, because the `tensorflow-datasets` package raised a protobuf version error in Colab. A stratified 70:15:15 split with seed 42 was created once and reused by all six notebooks; the six result files contain identical test labels (551 images).

Table 1: Class distribution of the fixed 70:15:15 split.

Class	Train	Validation	Test	Total
`daisy`	443	95	95	633
`dandelion`	629	134	135	898
`roses`	449	96	96	641
`sunflowers`	489	105	105	699
`tulips`	559	120	120	799
Total	2569	550	551	3670

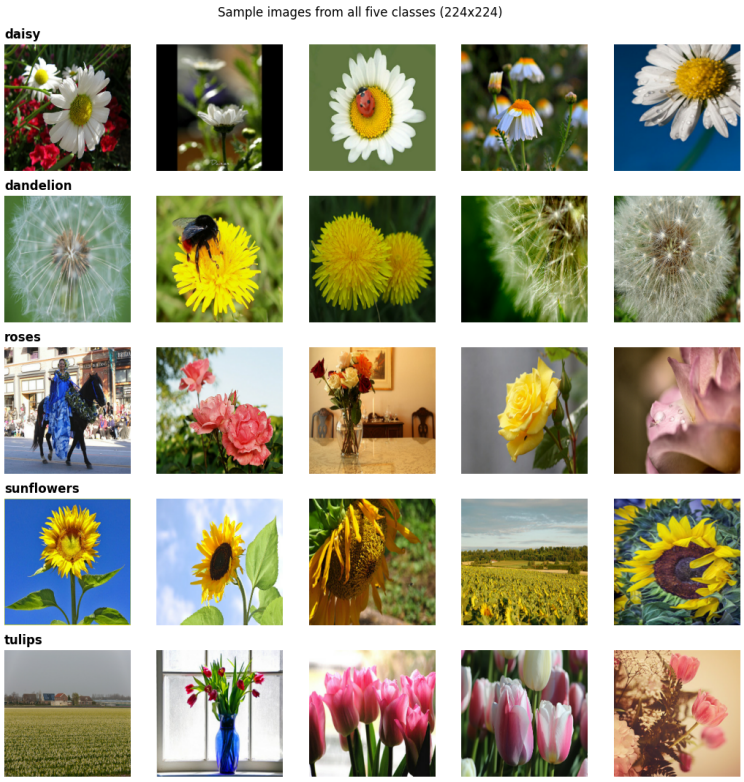


Figure 1: Sample images from all five classes (output of the data-loading task).

Inference: The classes differ in colour and shape, but the samples contain clutter and ambiguous cases (a horse parade labelled roses, distant tulip and sunflower fields); dandelion is the largest class (898 images) and daisy the smallest (633), so the data are mildly imbalanced.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	Spatial Resolution	Channels
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

4.1 Observed Feature Dimensions and Backbone Check

Table 2: Feature dimensions printed by the notebooks; they match the table above.

Position	**Layer (output of the stage)**	**Observed shape** (H, W, C)
P_1	conv2_block3_out	56, 56, 256
P_2	conv3_block4_out	28, 28, 512
P_3	conv4_block6_out	14, 14, 1024
P_4	conv5_block3_out	7, 7, 2048

Table 3: Backbone check printed in the architecture task (ImageNet weights copied into the rebuilt network).

Quantity	**A0**	**A1**	**A2**	**A3**	**A4**	**A5**
Max. |difference| at conv5 output vs. Keras ResNet50	0.00	32.97	39.00	50.23	0.00	50.23

The baseline has 23,597,957 parameters = 23,587,712 (ResNet50 without classifier) + 10,245 (Dense $2048 \to 5$).

Inference: The rebuilt network reproduces the pretrained ResNet50 exactly (difference 0.00 for A0 and A4, where A4's self-attention starts as an identity because $\gamma = 0$), so the weight transfer is correct; the other models differ because their attention block changes the features before conv5_x (A5 equals A3 since its self-attention branch also starts as an identity).

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

Which attention mechanism should be used?

and

Where should the attention mechanism be inserted?

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.

The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:
- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;

- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H \times W \times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\text{max}} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\text{max}}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H \times W \times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow F_{cs}}$$

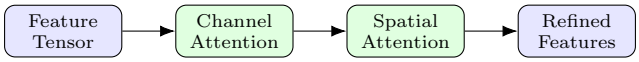
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\mathrm{SA}} = F + \gamma\,\mathrm{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

$$\boxed{\text{Which spatial features are related to other spatial features?}}$$

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

13.1 Observed Feature Dimensions Before and After the Attention Block

Table 4: Input and output shapes of the attention block `att_P3` printed by the notebooks.

Model	**Module (at P_3)**	**Input**	**Output**
A1	Channel attention	(14, 14, 1024)	(14, 14, 1024)
A2	Spatial attention (7×7)	(14, 14, 1024)	(14, 14, 1024)
A3	Channel $\rightarrow$ Spatial	(14, 14, 1024)	(14, 14, 1024)
A4	Self-attention	(14, 14, 1024)	(14, 14, 1024)
A5	Channel $\rightarrow$ Spatial $\rightarrow$ Self	(14, 14, 1024)	(14, 14, 1024)

Inference: Every module is shape-preserving, so conv5_x receives the same $14 \times 14 \times 1024$ tensor as in the baseline.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	Channel	Spatial	Self-Attention
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

14.1 Experimental Setup Used

Table 5: Training and evaluation protocol.

Item	Setting (identical for A0–A5)
Backbone / weights	ResNet50, ImageNet (Keras), input $224 \times 224 \times 3$, `resnet50.preprocess_input`
Split / seed	stratified 70:15:15 (2,569 / 550 / 551 images), seed 42
Augmentation	horizontal flip, rotation 0.05, zoom 0.10, translation 0.10, contrast 0.15
Head	Global average pooling, Dropout 0.3, Dense (5, softmax)
Phase 1 (10 epochs)	backbone frozen; attention + classifier trained; Adam, learning rate 10^{-3}
Phase 2 (10 epochs)	conv5_x unfrozen (BatchNorm kept frozen); Adam, learning rate 10^{-4}
Batch size / loss	32 / categorical cross-entropy
Attention settings	position P_3, reduction ratio $r = 16$, spatial kernel 7×7
Model for testing	weights of the best-validation epoch of phase 2
Inference time	ms per image, 160 test images, batch 32, mean of 3 passes after warm-up, Colab T4

14.2 Observed Results: Attention-Type Ablation at P_3

Table 6: Test results of the six models (macro precision, recall and F1; train/val accuracy of the saved best-validation checkpoint).

Model	Acc.	Prec.	Rec.	Macro F1	Total params	Trainable (fine-tune)	Train acc.	Val acc.	Infer. (ms/img)	Train time (min)
A0: Baseline ResNet50	0.9347	0.9385	0.9327	0.9347	23,597,957	14,963,717	0.9938	0.9418	7.10	14.1
A1: Channel Attention	0.9328	0.9316	0.9359	0.9329	23,730,117	15,095,877	0.9918	0.9400	7.68	15.3
A2: Spatial Attention	0.9437	0.9483	0.9405	0.9432	23,598,056	14,963,816	0.9961	0.9455	7.71	14.2
A3: Channel+Spatial	0.9383	0.9411	0.9379	0.9381	23,730,216	15,095,976	0.9879	0.9273	7.32	13.8
A4: Self-Attention	0.9328	0.9347	0.9314	0.9328	24,909,958	16,275,718	0.9942	0.9491	7.51	14.0
A5: Combined (C+S+SA)	0.9220	0.9210	0.9223	0.9214	25,042,217	16,407,977	0.9872	0.9418	7.96	15.9

Inference: All six models lie between 0.9220 (A5 Combined) and 0.9437 (A2 Spatial) accuracy; the baseline reaches 0.9347, so no attention type gives a clear gain.

Table 7: Sampling uncertainty (Wilson 95% interval) and paired exact McNemar test against the baseline, computed from the stored test predictions.

Model	Correct (of 551)	95% interval of accuracy	Baseline right / model right only	McNemar p vs. baseline
A0: Baseline ResNet50	515 / 551	91.1–95.2%	–	–
A1: Channel Attention	514 / 551	90.9–95.1%	20 / 19	1.000
A2: Spatial Attention	520 / 551	92.1–96.0%	11 / 16	0.442
A3: Channel+Spatial	517 / 551	91.5–95.6%	11 / 13	0.839
A4: Self-Attention	514 / 551	90.9–95.1%	17 / 16	1.000
A5: Combined (C+S+SA)	508 / 551	89.7–94.2%	24 / 17	0.349

Inference: The 95% interval is about ±2 points and every McNemar p value is at least 0.349, so none of the differences between models is statistically significant.

Table 8: Class-wise F1 on the test set (difference to the baseline in brackets).

Model	daisy	dandelion	roses	sunflowers	tulips	Macro F1
A0: Baseline ResNet50	0.958	0.952	0.902	0.961	0.900	0.935
A1: Channel Attention	0.938 (−0.020)	0.928 (−0.024)	0.921 (+0.019)	0.938 (−0.023)	0.940 (+0.039)	0.933
A2: Spatial Attention	0.952 (−0.006)	0.963 (+0.011)	0.917 (+0.015)	0.967 (+0.006)	0.917 (+0.017)	0.943
A3: Channel+Spatial	0.930 (−0.028)	0.951 (−0.001)	0.924 (+0.022)	0.966 (+0.005)	0.920 (+0.020)	0.938
A4: Self-Attention	0.941 (−0.017)	0.945 (−0.007)	0.902 (−0.001)	0.966 (+0.005)	0.911 (+0.010)	0.933
A5: Combined (C+S+SA)	0.947 (−0.011)	0.941 (−0.011)	0.891 (−0.011)	0.930 (−0.031)	0.897 (−0.003)	0.921

Inference: Roses and tulips are the weakest classes for every model; A1–A3 improve roses by +0.015 to +0.022 and tulips by +0.017 to +0.039, but every attention model loses 0.006 to 0.028 F1 on daisy.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	Attention Position	Resolution	Channels
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;

- intermediate representations; or
- low-resolution semantic features.

The principal question is:

How does the location of attention influence performance?

15.1 Observed Status

Table 9: Attention-position ablation.

Experiment	Status	Accuracy / macro F1
B1: after conv2_x (P_1)	not run	–
B2: after conv3_x (P_2)	not run	–
B3: after conv4_x (P_3)	= model A3	0.9383 / 0.9381
B4: after conv5_x (P_4)	not run	–

Inference: Only the P_3 model (B3 = A3) was trained, so this run gives no evidence about the effect of attention position.

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

Is attention at more locations necessarily more effective?

16.1 Observed Status

Table 10: Multiple attention positions.

Configuration	Status	Accuracy / macro F1
Attention only at P_3	= model A3	0.9383 / 0.9381
Attention only at P_4	not run	–
Attention at $P_3 + P_4$	not run	–
Attention at $P_2 + P_3 + P_4$ (extended)	not run	–

Inference: Multi-stage attention was not trained; the question whether more locations are more effective cannot be answered from this run.

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

17.1 Observed Status

Table 11: Attention order.

Order	Status	Accuracy / macro F1
Channel → Spatial at P_3	= model A3	0.9383 / 0.9381
Spatial → Channel at P_3	not run	–

Inference: Only Channel → Spatial was trained, so the two orders cannot be compared.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	Accuracy	Macro F1	Inference Time
3×3	–	–	–
5×5	–	–	–
7×7	0.9383	0.9381	7.32 ms

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

18.1 Observed Status

Inference: The table above contains only the 7×7 kernel (model A3); the 3×3 and 5×5 models were not run, so the effect of kernel size is not evaluated.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.

Students should select at least

$$\boxed{\text{two additional attention mechanisms}}$$

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

$$\boxed{\text{ResNet50}}$$

for all comparisons.

To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	**Accuracy**	**Precision**	**Recall**	**Macro F1**	**Parameters**	**Inference Time**
No Attention (A0)	0.9347	0.9385	0.9327	0.9347	23,597,957	7.10 ms
Channel / SE (A1)	0.9328	0.9316	0.9359	0.9329	23,730,117	7.68 ms
Spatial Attention (A2)	0.9437	0.9483	0.9405	0.9432	23,598,056	7.71 ms
ECA (not run)	–	–	–	–	–	–
CBAM (not run)	–	–	–	–	–	–
Self-Attention (A4)	0.9328	0.9347	0.9314	0.9328	24,909,958	7.51 ms
Channel+Spatial (A3, own implementation)	0.9383	0.9411	0.9379	0.9381	23,730,216	7.32 ms
Combined C+S+SA (A5)	0.9220	0.9210	0.9223	0.9214	25,042,217	7.96 ms

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. How does classification accuracy vary across attention mechanisms?
2. How does macro F1-score vary across attention mechanisms?
3. How many parameters are introduced by each attention module?
4. How does each attention mechanism influence inference time?
5. Does attention improve the convergence behavior?
6. Does attention produce more focused feature localization?
7. Does an increase in attention complexity always improve classification performance?
8. Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

19.3 Observed Status and Answers for the Models Run

ECA and CBAM (and any other established mechanism) were not trained, so the requirement of at least two additional mechanisms is **not yet fulfilled**. A1 uses the squeeze-and-excitation formulation (global average pooling, bottleneck $r = 16$, sigmoid) and therefore serves as the Channel/SE row; A3 is a sequential channel–spatial module of the CBAM type, but it is the lab's own implementation (CBAM's channel branch also uses max pooling).

1. **Accuracy:** 0.9220–0.9437 across the six models; none differs significantly from the baseline.
2. **Macro F1:** 0.9214–0.9432, in the same order as accuracy.
3. **Parameters added:** A1 132,160; A2 99; A3 132,259; A4 1,312,001; A5 1,444,260.
4. **Inference time:** +0.22 to +0.86 ms per image over the baseline (7.10 ms); the order is not monotonic with module size (A3 is faster than A1 and A2), so differences below about 1 ms are not reliable.
5. **Convergence:** A4 and A5 reach 0.90 validation accuracy at epochs 2 and 3, against epoch 4 (baseline), 5 (A1) and 6 (A2, A3); after fine-tuning all models end at similar validation accuracy.
6. **Focused localisation:** Grad-CAM compactness is higher than the baseline (0.496) for A2 (0.549) and A3 (0.588) and lower for A1 (0.454), A4 (0.422) and A5 (0.401).
7. **Does complexity always help?** No: A5, the most complex model, has the lowest accuracy (0.9220).
8. **Lightweight versus heavy:** Yes: A2 adds only 99 parameters and has the highest accuracy (0.9437), above A4 (+1,312,001 parameters, 0.9328).

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.
8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.

The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.

For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.

The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Students should summarize the major experiments using the following table.

Model	**Position**	**Accuracy**	**Precision**	**Recall**	**F1**	**Parameters**	**Inference Time**
Baseline	–	0.9347	0.9385	0.9327	0.9347	23,597,957	7.10 ms
Channel Attention	P_3	0.9328	0.9316	0.9359	0.9329	23,730,117	7.68 ms
Spatial Attention	P_3	0.9437	0.9483	0.9405	0.9432	23,598,056	7.71 ms
Channel+Spatial	P_2	–	–	–	–	–	–
Channel+Spatial	P_3	0.9383	0.9411	0.9379	0.9381	23,730,216	7.32 ms
Channel+Spatial	P_4	–	–	–	–	–	–
Self-Attention	P_3	0.9328	0.9347	0.9314	0.9328	24,909,958	7.51 ms
Additional Attention-1	P_3	–	–	–	–	–	–
Additional Attention-2	P_3	–	–	–	–	–	–

Inference: The P_3 rows are filled from Table 6; the P_2, P_4 and additional-attention rows are marked "–" because those models were not trained.

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

Epoch versus Accuracy

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.

Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

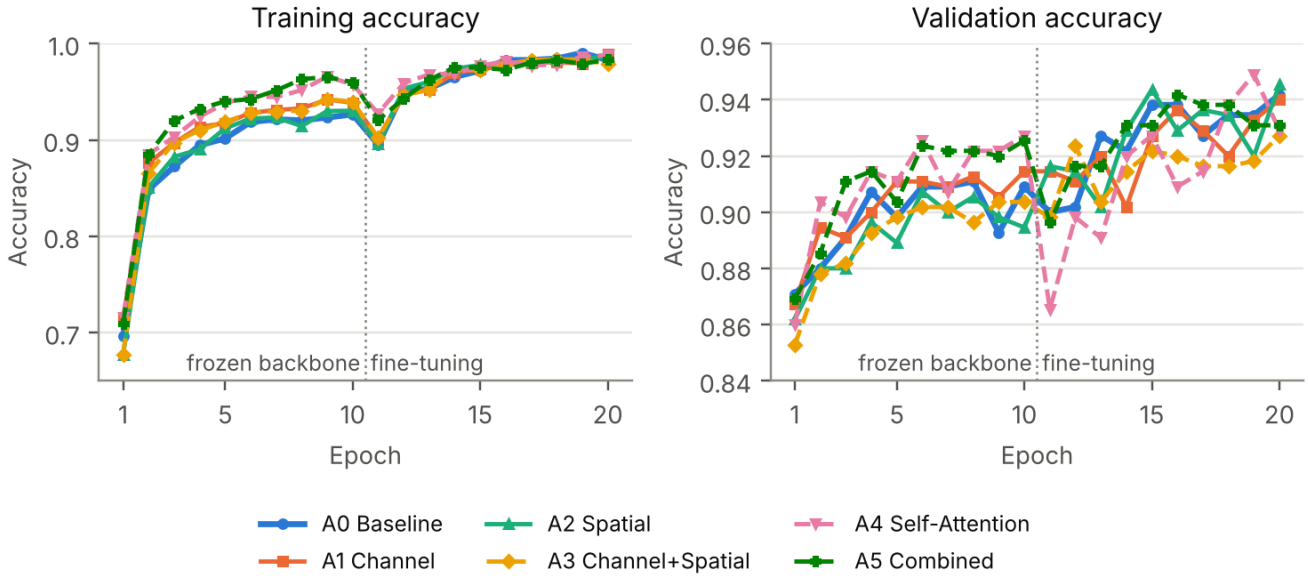
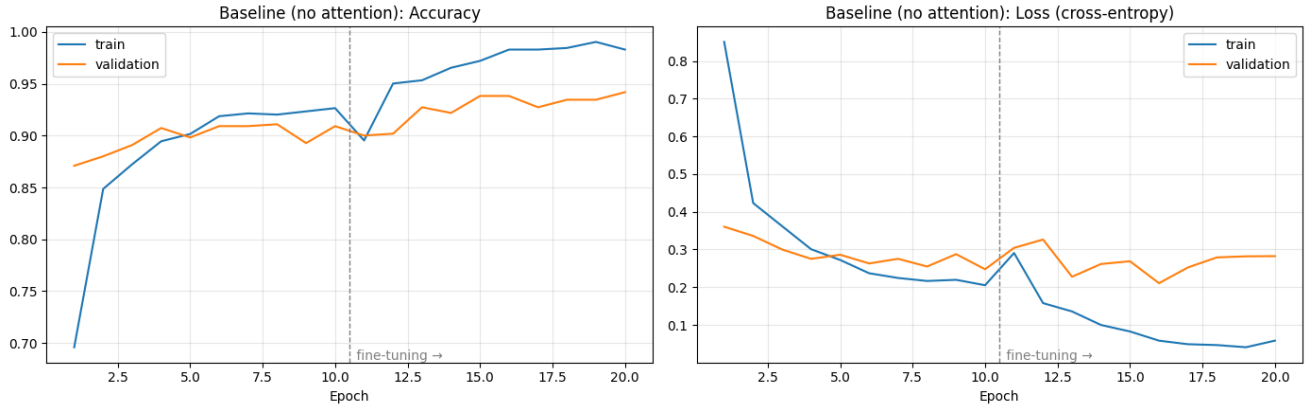
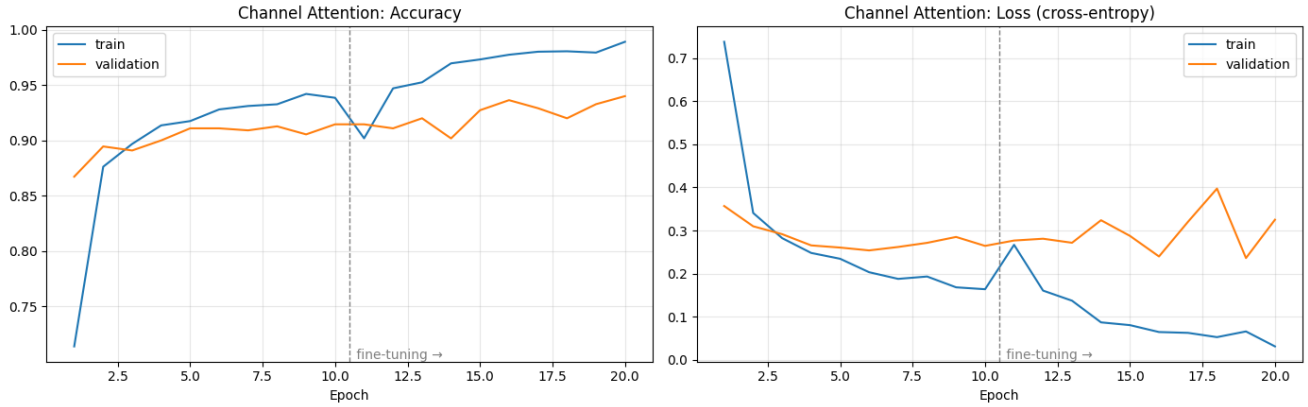


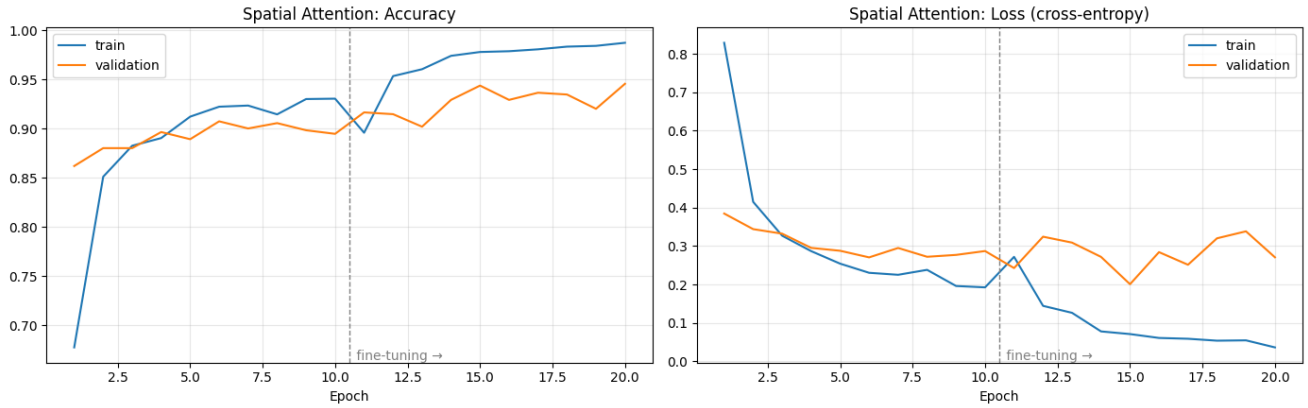
Figure 2: Training and validation accuracy versus epoch for all six models.



(a) A0: Baseline ResNet50

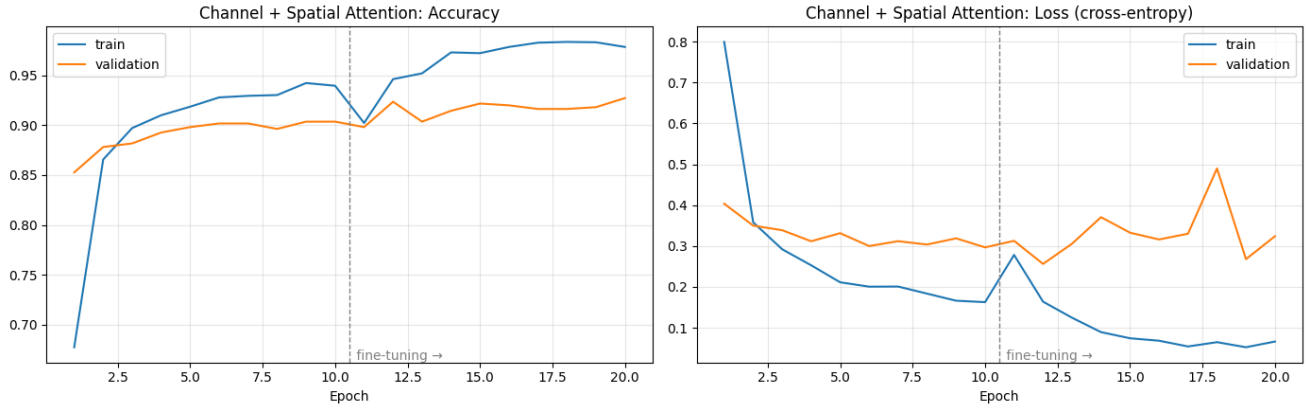


(b) A1: Channel Attention

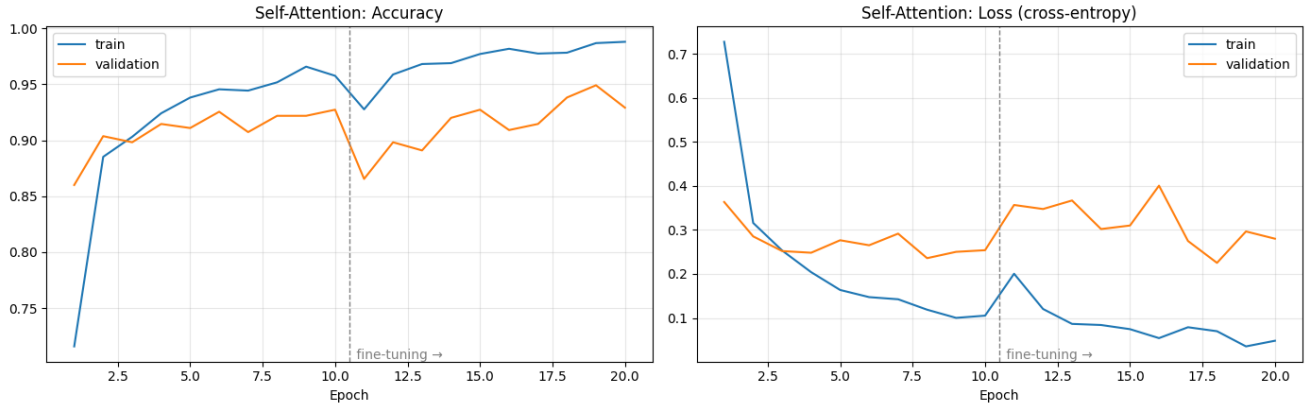


(c) A2: Spatial Attention

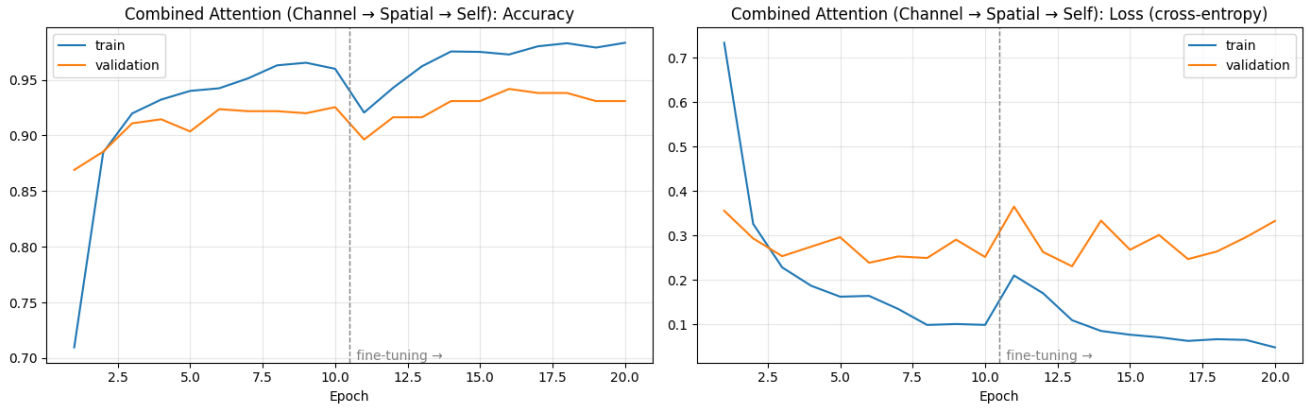
Figure 3: Notebook outputs (accuracy and loss curves) of A0–A2.



(a) A3: Channel+Spatial



(b) A4: Self-Attention



(c) A5: Combined (C+S+SA)

Figure 4: Notebook outputs (accuracy and loss curves) of A3–A5.

Inference: Validation accuracy rises from about 0.86–0.87 in epoch 1 to 0.89–0.93 at the end of the frozen phase and to 0.927–0.949 after fine-tuning, while training accuracy reaches 0.98–0.99.

Inference: A4 and A5 converge fastest in the frozen phase (validation ≥ 0.90 at epochs 2 and 3 versus 4 for the baseline), but they also train 1,322,246–1,454,505 parameters in that phase against 10,245.

Inference: Unfreezing conv5_x lowers training accuracy by 3.0–3.9 points in every model, and validation accuracy fluctuates by 3–8 points over the fine-tuning epochs (A4 the most, 8.4), which is as large as the differences between models.

Inference: Attention does not close the train–validation gap: at epoch 20 the gap is 4.1 points for the baseline and 4.2–5.9 for the attention models, so the baseline has the smallest gap and no model is unstable.

24.2 Training and Validation Loss

Plot

Epoch versus $\mathcal{L}_{CE}$.

The graph should include training and validation loss.
Students should compare the convergence behavior of the baseline and attention models.

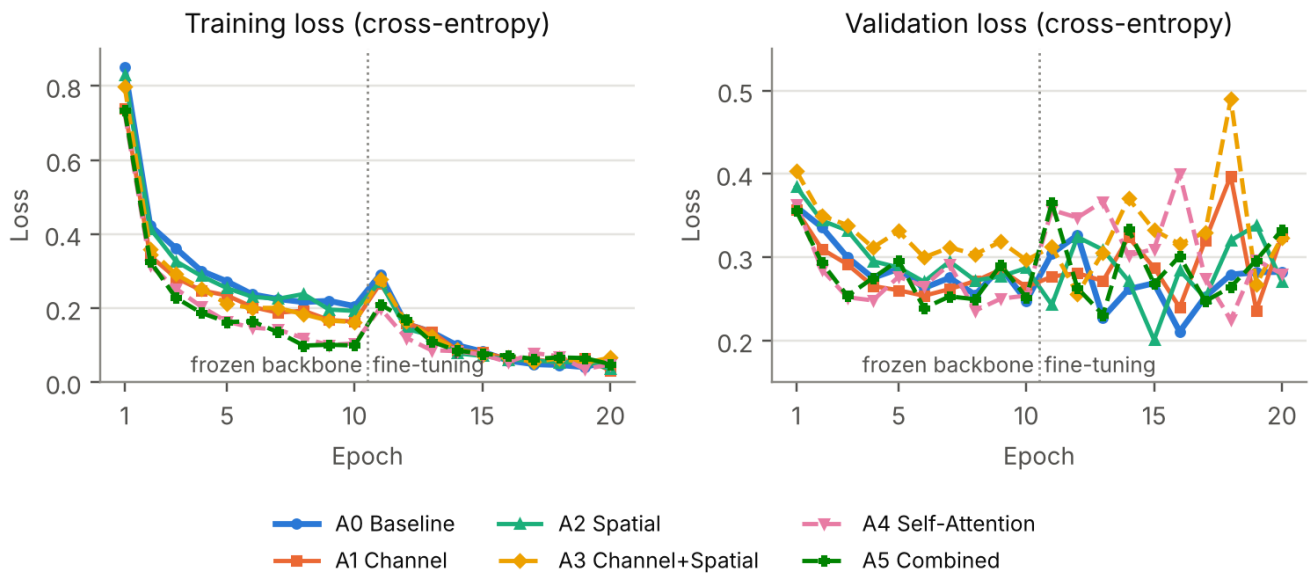


Figure 5: Training and validation loss (categorical cross-entropy) versus epoch for all six models.

Table 12: Convergence statistics read from the training histories (training accuracy is measured with augmentation).

Model	Val acc. ep. 10	First epoch val ≥ 0.90	Best val acc. (epoch)	Train acc. ep. 20	Val acc. ep. 20	Gap (pts)	Min val loss (epoch)	Val loss ep. 20	Train loss ep. 20
A0: Baseline ResNet50	0.909	4	0.942 (20)	0.983	0.942	4.1	0.211 (16)	0.282	0.058
A1: Channel Attention	0.915	5	0.940 (20)	0.989	0.940	4.9	0.236 (19)	0.325	0.031
A2: Spatial Attention	0.895	6	0.945 (20)	0.987	0.945	4.2	0.201 (15)	0.271	0.036
A3: Channel+Spatial	0.904	6	0.927 (20)	0.979	0.927	5.1	0.256 (12)	0.324	0.066
A4: Self-Attention	0.927	2	0.949 (19)	0.988	0.929	5.9	0.225 (18)	0.280	0.048
A5: Combined (C+S+SA)	0.925	3	0.942 (16)	0.983	0.931	5.2	0.231 (13)	0.333	0.049

Inference: Training loss falls to 0.03–0.07, but validation loss bottoms out at 0.20–0.26 (epochs 12–19) and ends at 0.27–0.33, i.e. all models begin to overfit during fine-tuning.

Inference: The three models that contain channel attention (A1, A3, A5) end with the highest validation loss (0.325, 0.324, 0.333) compared with 0.282 for the baseline.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

$$\text{Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM}.$$

24.4 Attention Mechanism versus Macro F1

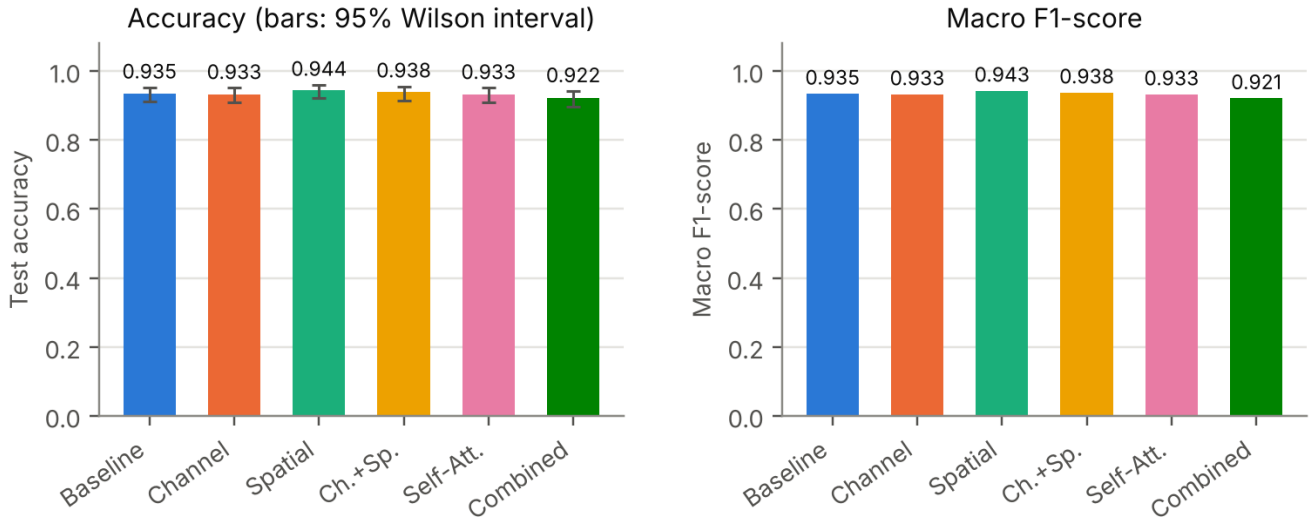
Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score}.$$

This comparison is useful when the class-wise performance is not uniform.



(a) Test accuracy

(b) Macro F1-score

Figure 6: Classification performance of the six attention configurations (bars start at zero; whiskers: 95% Wilson interval).

Inference: The bars differ by at most 2.2 points, less than the width of the 95% intervals, and macro F1 gives the same order as accuracy (A2, A3, A0, A1, A4, A5): the ranking is not statistically reliable.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy}.$$

Each point should represent one attention mechanism.
This allows the student to study the

$$\boxed{\text{Accuracy–Complexity Trade-off}}$$

instead of evaluating accuracy independently.

24.6 Macro F1 versus Inference Time

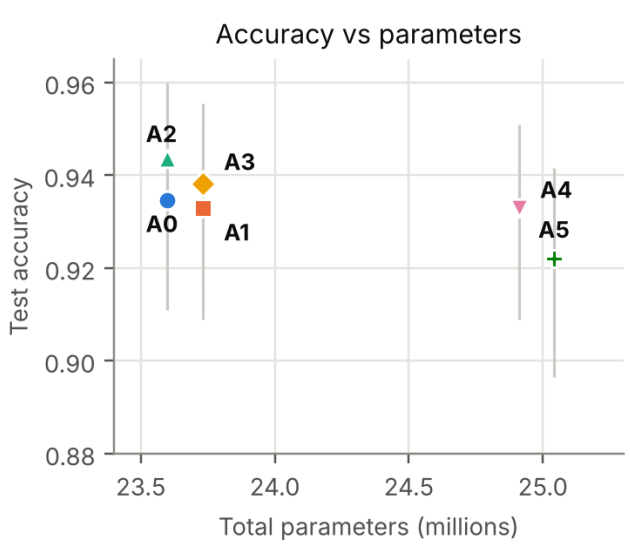
Plot:

$$x = \text{Inference Time per Image}$$

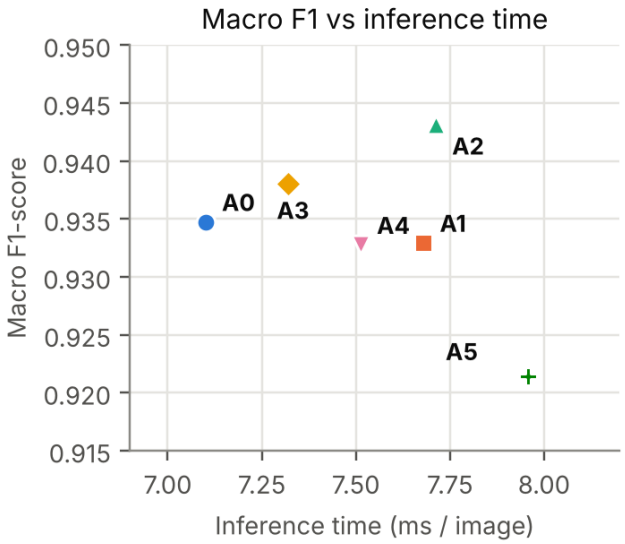
and

$$y = \text{Macro F1-score}.$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.



(a) Accuracy versus total parameters (grey bars: 95% interval)



(b) Macro F1 versus inference time

Figure 7: Accuracy–complexity and F1–latency trade-offs of the six models (A0–A5 as in the tables).

Inference: Accuracy does not rise with parameter count: A2 adds 99 parameters and scores highest, whereas A4 and A5 (+1.31 M and +1.44 M) score at or below the baseline.

Inference: A5 is the slowest (7.96 ms) with the lowest F1, and A2 gains 0.0084 F1 for +0.61 ms; since A3 is faster than A1 and A2, latency differences below 1 ms are not reliable.

25. Confusion Matrix Analysis

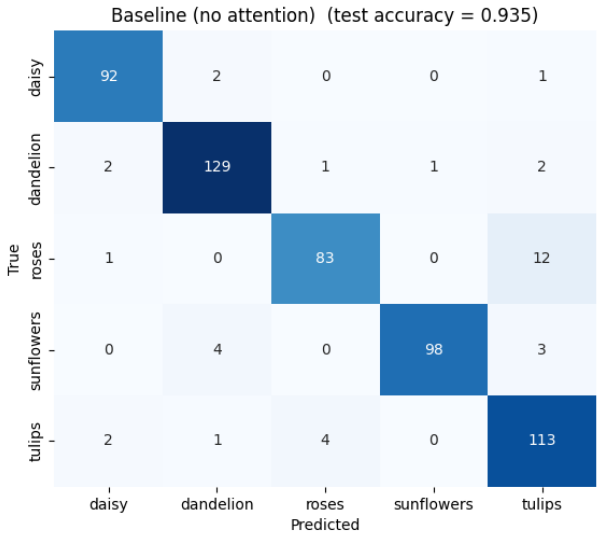
Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

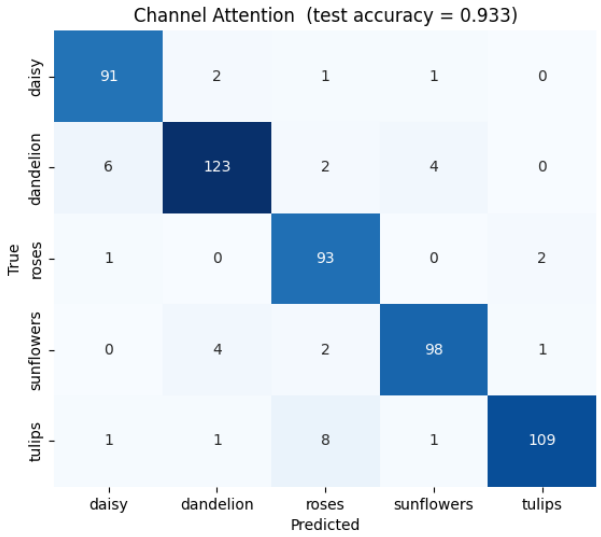
The confusion matrix should contain all five flower classes.

Students should analyze:

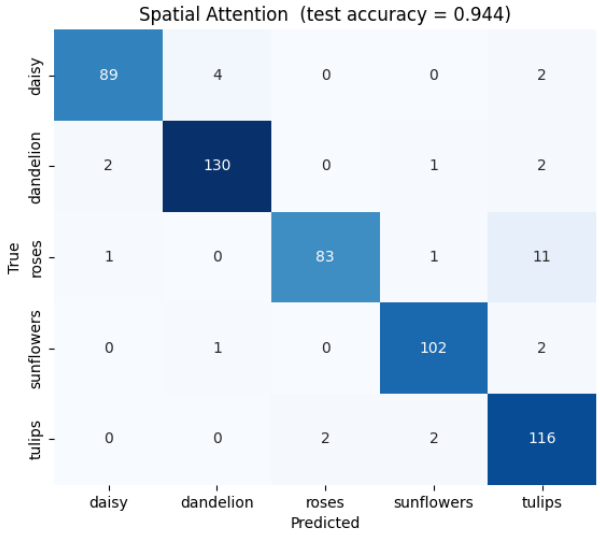
- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.



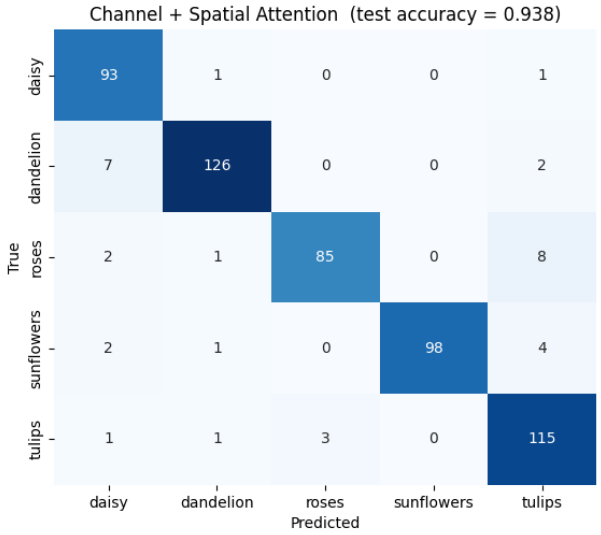
(a) A0: Baseline ResNet50



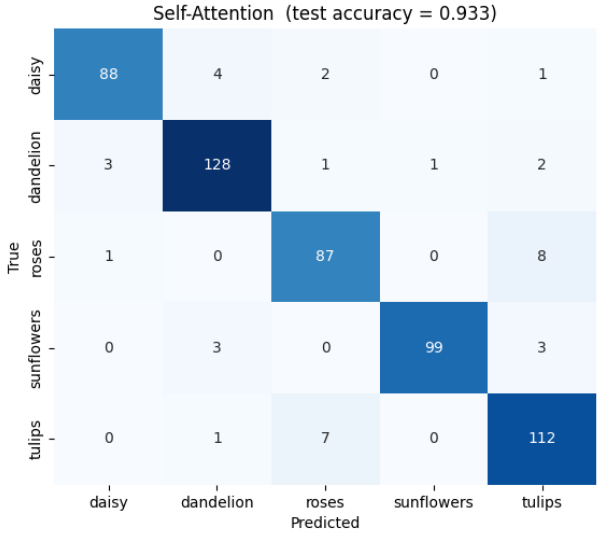
(b) A1: Channel Attention



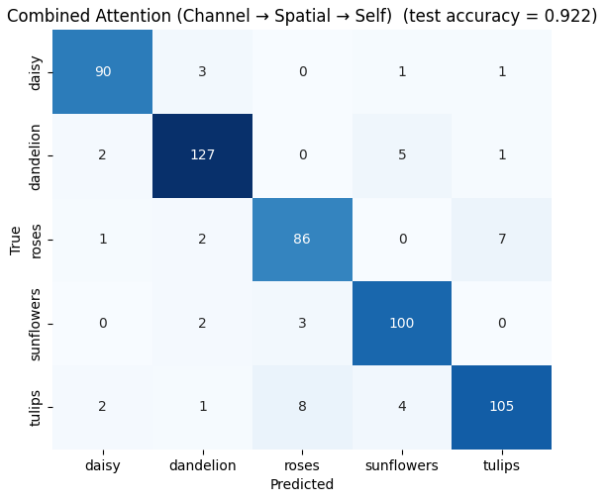
(c) A2: Spatial Attention



(d) A3: Channel+Spatial



(e) A4: Self-Attention



(f) A5: Combined (C+S+SA)

Figure 8: Confusion matrices on the 551 test images (notebook outputs). The baseline (A0) and the attention models A1–A5 are shown side by side.

Table 13: Most frequent confusions, counted from the confusion matrices.

Model	roses → tulips	tulips → roses	roses↔tulips total	dandelion → daisy	daisy → dandelion
A0: Baseline ResNet50	12	4	16	2	2
A1: Channel Attention	2	8	10	6	2
A2: Spatial Attention	11	2	13	2	4
A3: Channel+Spatial	8	3	11	7	1
A4: Self-Attention	8	7	15	3	4
A5: Combined (C+S+SA)	7	8	15	2	3

Inference: The most accurate classes are sunflowers, daisy and dandelion (F1 up to 0.967), and roses ↔ tulips is the dominant confusion for all six models (16 errors for the baseline).

Inference: Attention moves errors instead of removing them: A1 cuts roses→tulips from 12 to 2 but raises tulips→roses from 4 to 8 and dandelion→daisy from 2 to 6; the combined roses/tulips errors fall only to 10–15.

Inference: Errors are largely shared: 79 of 551 images are misclassified by at least one model, 41 by at least three and 6 by all six, so the models differ only on a small set of borderline images.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input	Baseline Grad-CAM	Attention Model Grad-CAM

for the same image.

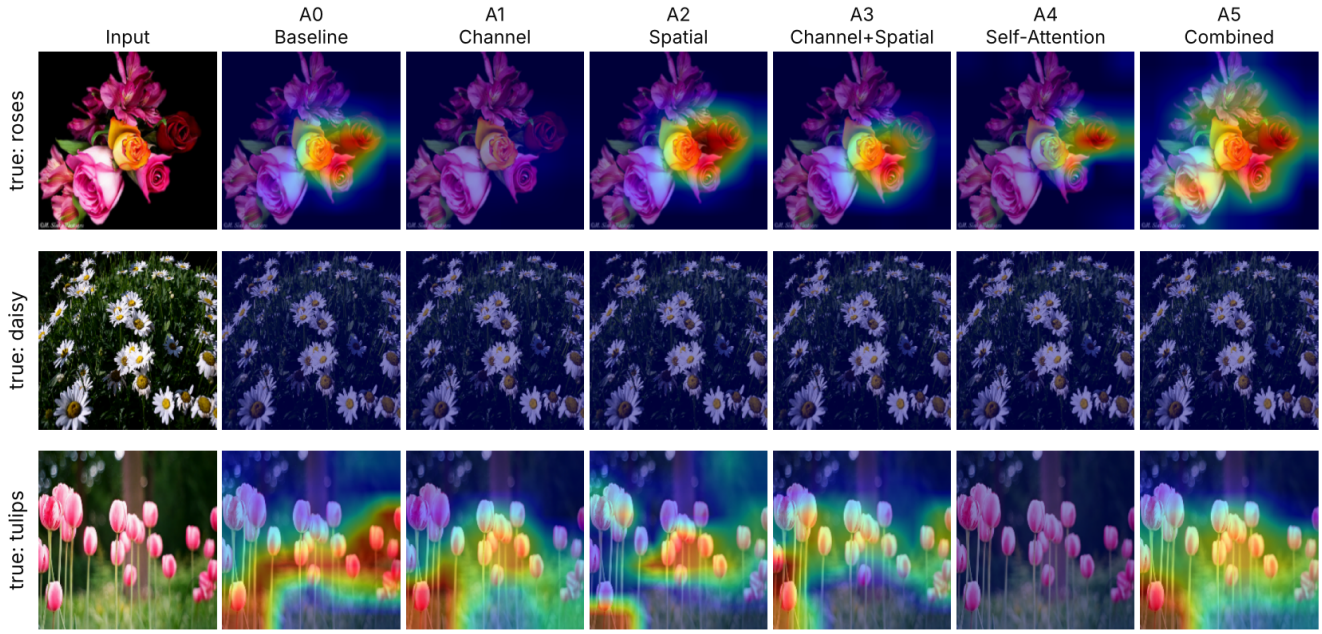


Figure 9: Grad-CAM (last convolutional stage, class score = softmax probability of the predicted class) for the same three test images and all six models. All six models predict the true class with probability 1.00 for these images. Panels are cropped from the notebook outputs.

Inference: Daisy (row 2): all six maps are almost empty although the prediction is 1.00, because the softmax probability saturates and its gradient vanishes, so Grad-CAM gives no localisation for very confident predictions.

Inference: Bouquet (row 1): the baseline and A2 highlight the orange and dark-red roses, A3 the orange and the lower-right pink rose, A4 only the dark-red rose, A5 spreads over all roses, and A1 shows almost no activation.

Inference: Tulips (row 3): the visible maps cover the blooms and also the stems and grass in the lower half (A4 is nearly blank), so attention does not remove background evidence.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

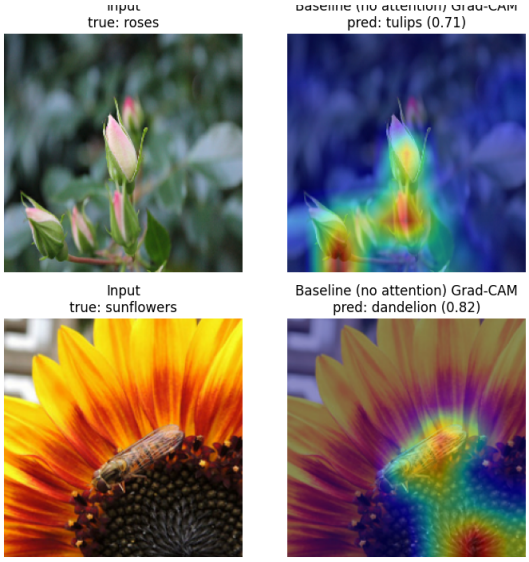
- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

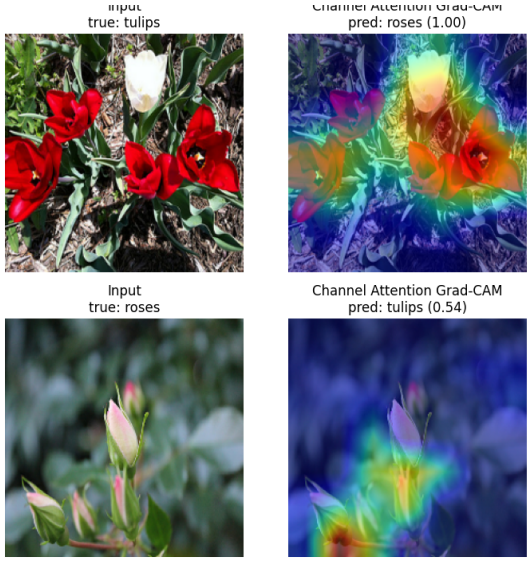
> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

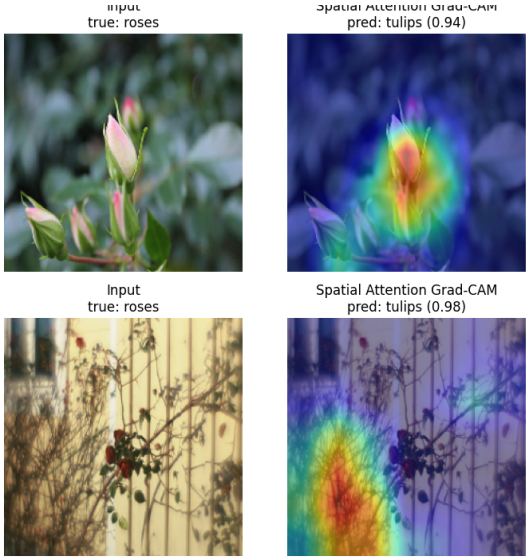
> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.



(a) A0: Baseline ResNet50



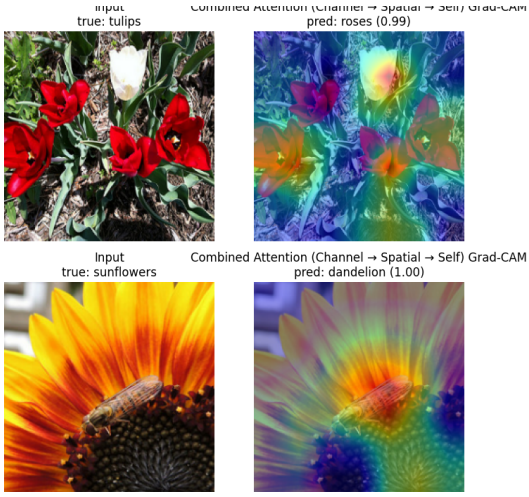
(b) A1: Channel Attention



(c) A2: Spatial Attention

(d) A3: Channel+Spatial

(e) A4: Self-Attention



(f) A5: Combined (C+S+SA)

Figure 10: Grad-CAM of two misclassified test images per model (notebook outputs; each model shows its own first two errors).

Inference: For the rose bud (true class roses) A0–A4 all predict tulips (confidence 0.54–0.97) with the activation lying on the bud itself, and only A5 classifies it correctly.

Inference: For the red tulip scene with a white tulip, A1, A3, A4 and A5 predict roses (0.91–1.00) with strong activation on the white tulip, whereas A0 and A2 are correct; for A2's error (roses among branches, predicted tulips 0.98) the activation falls on the branch area instead of a clear flower.

Inference: The errors are therefore confident class confusions on the flower region, not obvious mislocalisation onto the background.

Table 14: Share of the total Grad-CAM activation inside the 20% most active pixels, averaged over the first 30 test images (a blank map counts as 0).

Model	**Grad-CAM compactness**
A0: Baseline ResNet50	0.496
A1: Channel Attention	0.454
A2: Spatial Attention	0.549
A3: Channel+Spatial	0.588
A4: Self-Attention	0.422
A5: Combined (C+S+SA)	0.401

Inference: A2 (0.549) and A3 (0.588) give more concentrated maps than the baseline (0.496), while A1, A4 and A5 give more diffuse or blank maps (0.454, 0.422, 0.401); this is based on 30 images and is only indicative.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

$$\boxed{\text{Internal Attention}}$$

and

$$\boxed{\text{Post-hoc Explanation}}$$

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or
- channel weighting changes substantially for different input images.

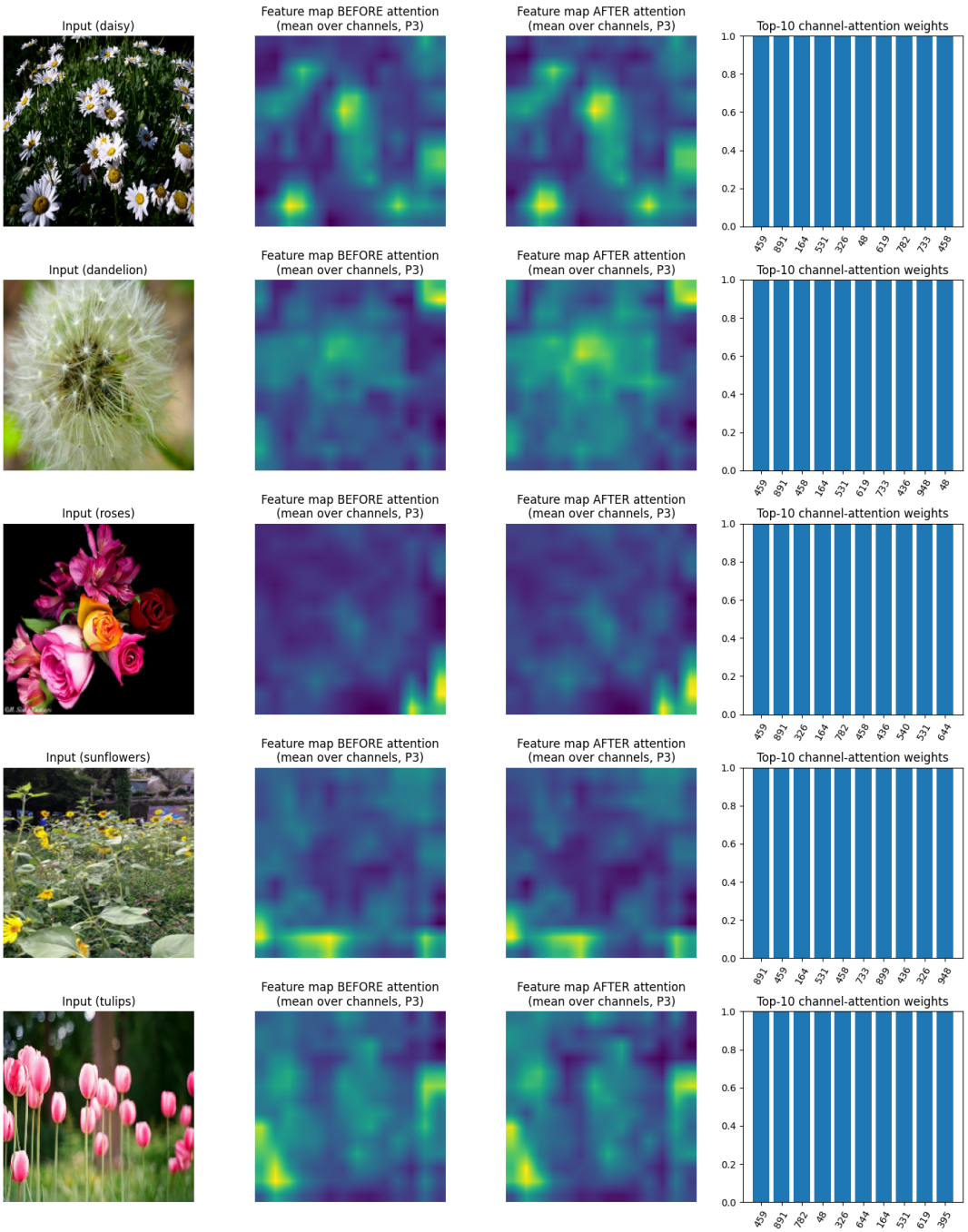


Figure 11: Channel attention (A1): input, mean feature map before attention, mean feature map after attention, and the top-10 channel-attention weights for one test image per class (notebook output).

Inference: The ten largest channel weights are all about 1.0, and four channels (459, 891, 164, 531) are among them for all five images, so the channel weighting is nearly image-independent.

Inference: Weights are close to binary (mean 0.593, std across channels 0.407, near the maximum possible 0.49 for this mean) and 56% exceed 0.7: about half of the channels pass unchanged rather than a few dominating, and the mean feature maps before and after attention look almost identical.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.

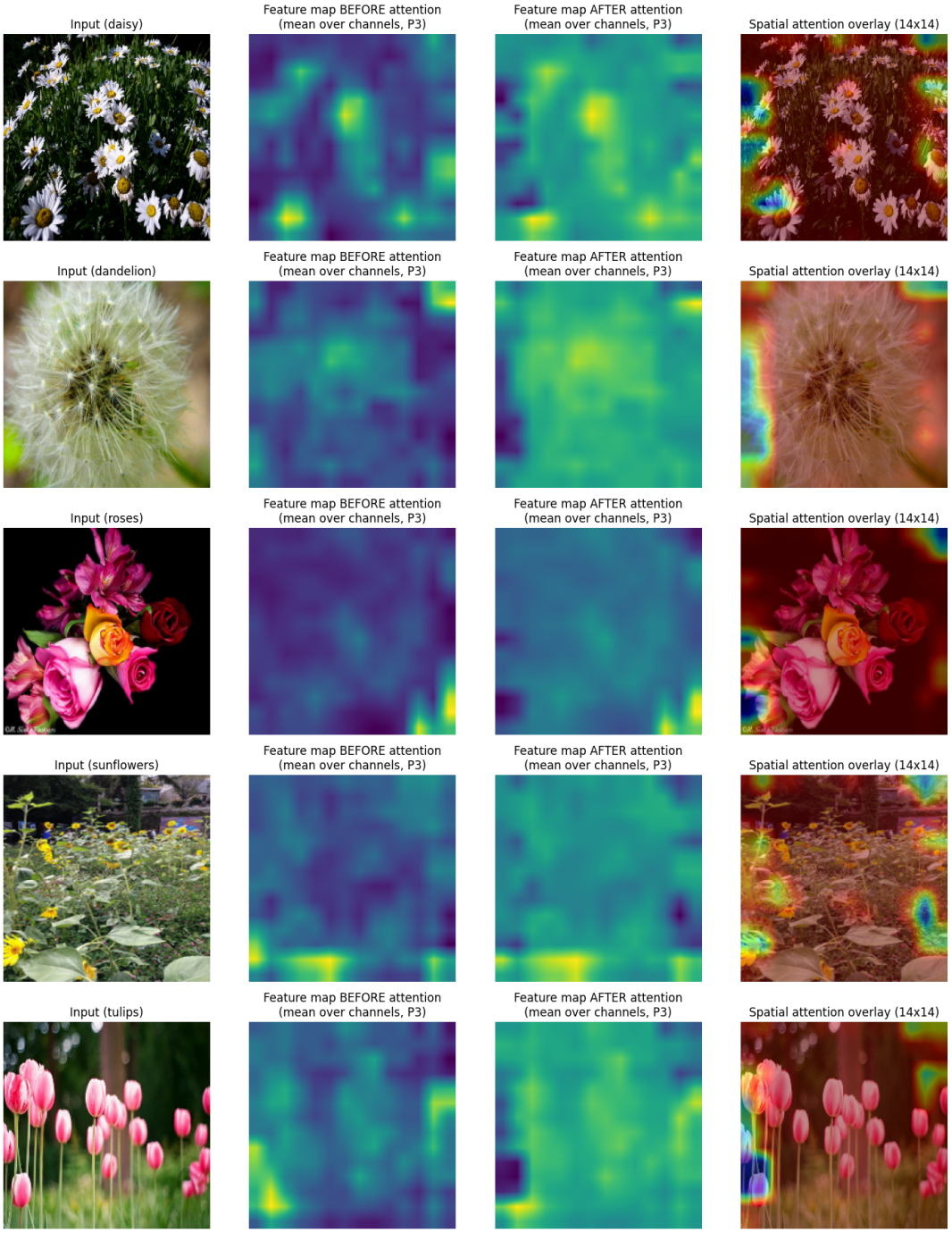


Figure 12: Spatial attention (A2): input, mean feature map before and after attention, and the spatial-attention overlay (14×14 map resized to 224×224, min–max normalised per image; blue = low, red = high).

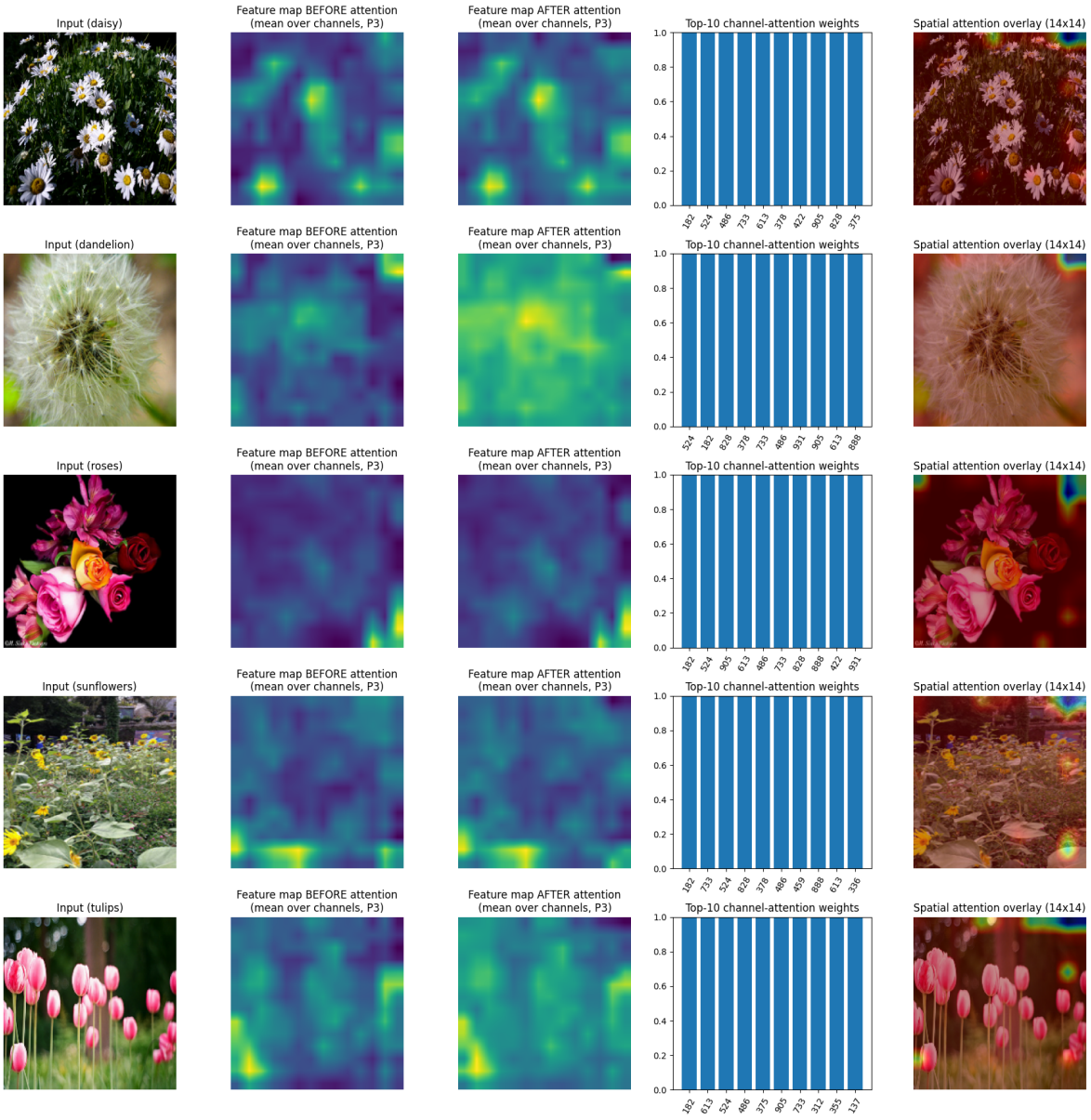


Figure 13: Channel → Spatial attention (A3): feature maps before and after attention, top-10 channel weights and spatial-attention overlay.

Inference: Most of the image receives a high spatial weight (dark red); low weights (blue) appear only in small patches near the image borders, so the module mostly suppresses borders and does not isolate the flower.

Inference: A3 shows the same two behaviours as A1 and A2 combined: near-binary channel weights (mean 0.599, 57% above 0.7) and border-dominated spatial maps, with only minor changes between the feature maps before and after attention.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution. Students should analyze how distant regions interact within the feature map.

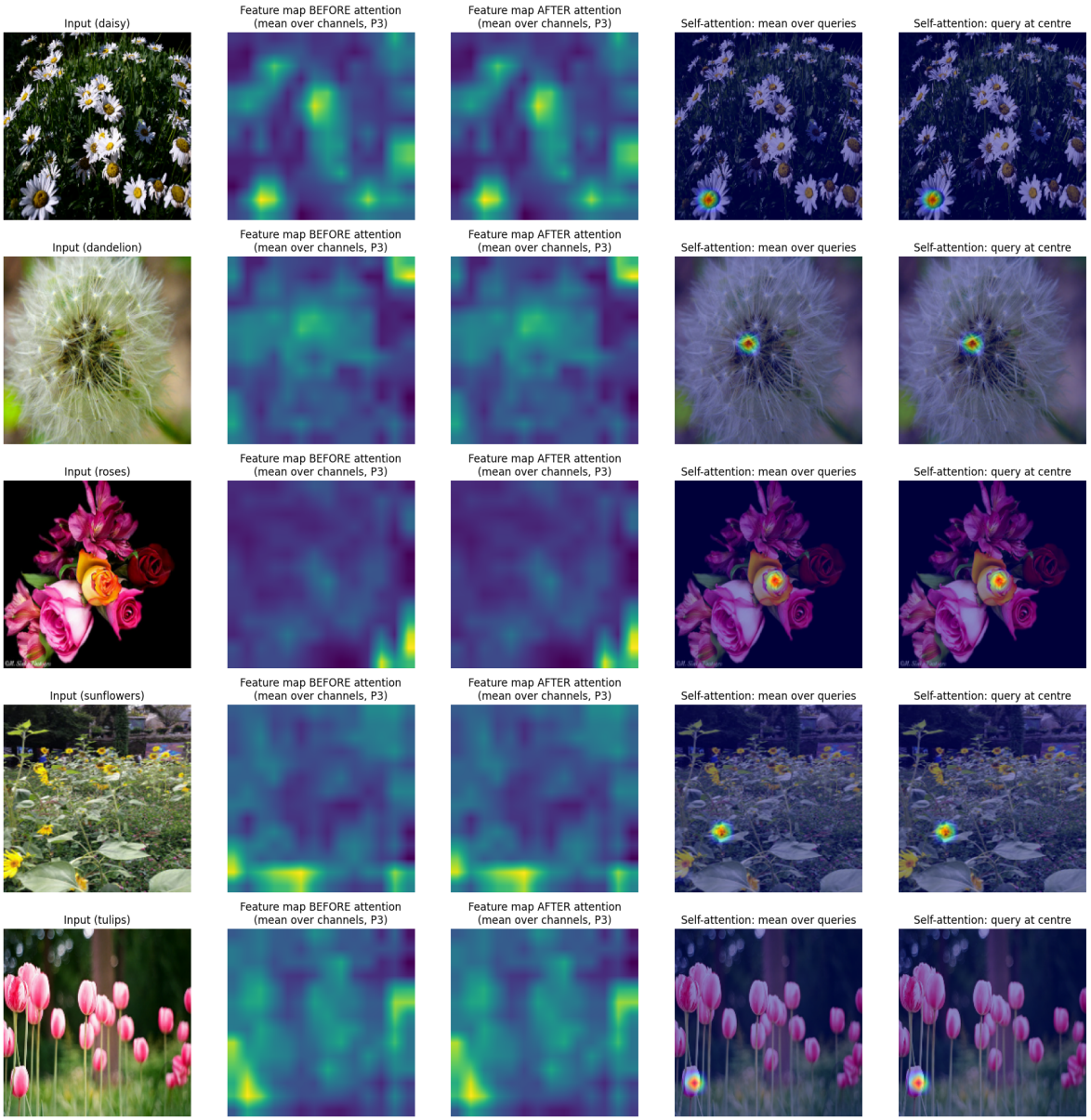


Figure 14: Self-attention (A4): feature maps before and after attention, mean attention over all query positions and attention of the centre query (14×14 maps resized and overlaid).

Inference: The mean-over-queries map and the centre-query map are almost identical single compact hot spots on a flower or flower centre, i.e. all queries attend to nearly the same few positions, which behaves like global context pooling rather than local interaction between neighbours.

Inference: The learned residual scale is small ($\gamma = -0.063$ for A4, -0.058 for A5), so the feature maps before and after self-attention are nearly identical.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

$$\boxed{\text{Input} \mid M_s^{P_2} \mid M_s^{P_3} \mid M_s^{P_4}}$$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

29.1 Observed Status

Inference: Spatial attention at P_2 and P_4 (models B2, B4) was not trained, so $M_s^{P_2}$, $M_s^{P_3}$ and $M_s^{P_4}$ cannot be compared; the P_3 maps are shown in Section 28.2.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

3 correctly classified images

and

2 incorrectly classified images.

For every selected image, compare the behavior of multiple models.
A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

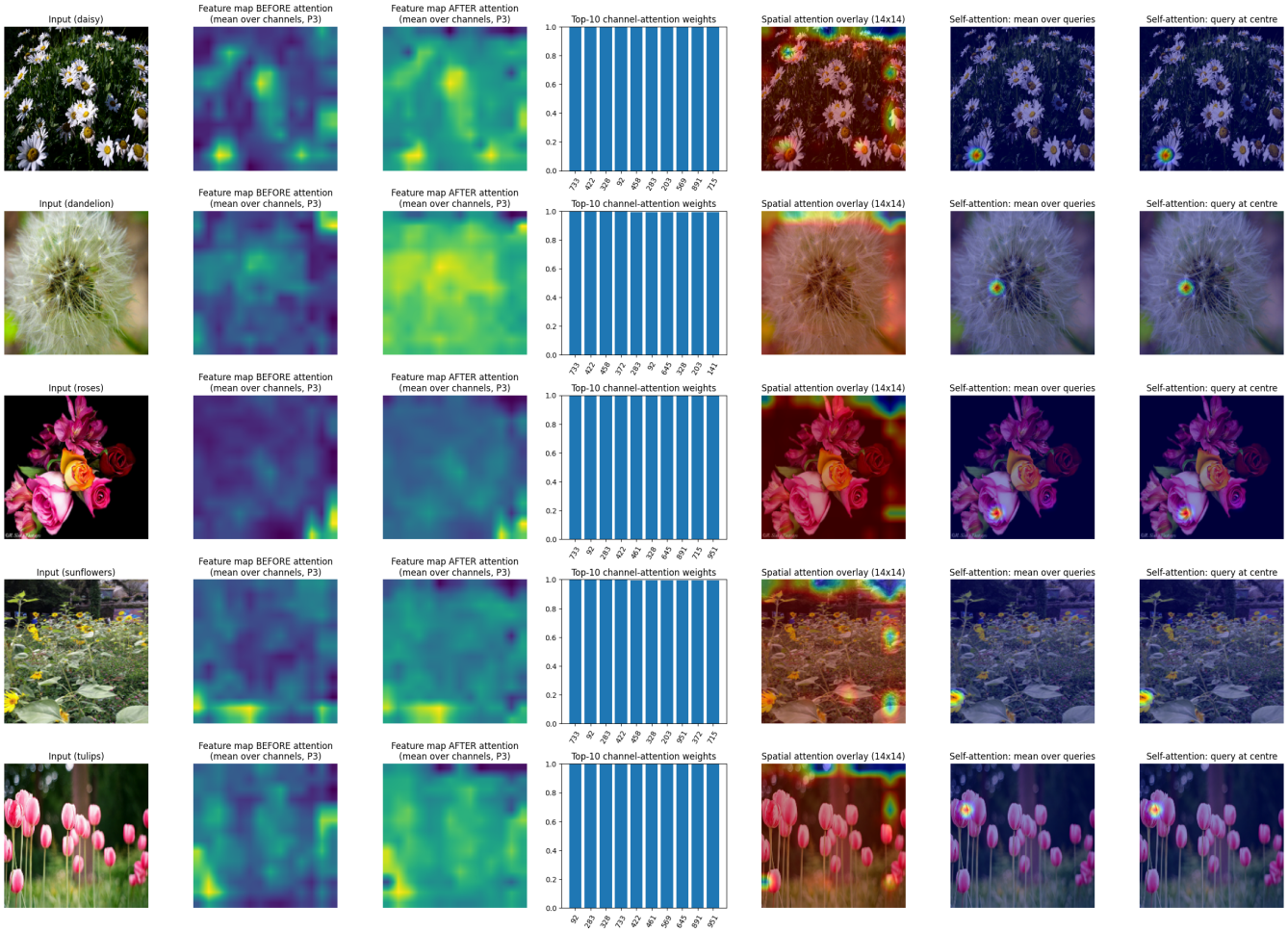


Figure 15: Combined attention (A5, Channel → Spatial → Self): feature maps before and after attention, top-10 channel weights, spatial-attention overlay and self-attention responses (notebook output).

Inference: The combined module shows the same three behaviours as the individual modules (saturated channel weights, border-dominated spatial map, single hot spot in self-attention), so stacking them adds parameters without adding new visible behaviour.

The Grad-CAM comparison for the same test images and the error cases are shown in Figures 9 and 10.

30.1 Qualitative Criteria

The student should discuss:

- foreground localization;
- background suppression;
- localization of petals;
- localization of the flower center;
- activation compactness;
- activation coverage;
- incorrectly attended regions;
- differences between correctly and incorrectly classified samples;
- differences between shallow and deep attention; and
- consistency of attention across different images.

Table 15: Summary of the qualitative criteria of Section 30.1.

Criterion	Observation
Foreground localisation	Visible Grad-CAM maps lie on the roses (bouquet) and tulip blooms in all models; no model concentrates on the flower alone.
Background suppression	Not observed: stems and grass remain active in the tulip image, and A2 activates on branches in an error case; the spatial gate is high almost everywhere.
Petals / flower centre	Maps for the rose bud and the white tulip sit on the flower body; fine petal or centre localisation cannot be resolved at the 7×7 Grad-CAM resolution.
Compactness and coverage	A2 and A3 are more compact than the baseline, A5 covers the widest area on the bouquet, A1 and A4 often give faint or blank maps (Table 14).
Incorrectly attended regions	Mostly the flower itself (rose bud, white tulip); background only in the A2 branch example.
Correct versus incorrect samples	Errors are confident (up to 1.00) with activation on the flower, so the maps differ mainly in the predicted class, not in location.
Shallow versus deep attention	Not evaluated: only P_3 was trained.
Consistency across images	Channel weights are nearly image-independent, spatial maps always lower the borders, and self-attention always finds one flower hot spot.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;
2. training and validation accuracy curves;
3. training and validation loss curves;
4. confusion matrix;
5. feature map before attention;
6. channel-attention weights;
7. spatial-attention heatmap;
8. spatial-attention overlay;
9. self-attention response for the self-attention experiment;
10. feature map after attention;
11. baseline Grad-CAM;
12. Grad-CAM for at least one attention-based model;
13. attention maps at P_2, P_3, and P_4; and
14. at least one graphical comparison of the performances of different attention mechanisms.

Table 16: Completion status of the required visualisations.

No.	Required visualisation (Section 31)	Status
1	Sample images of all five classes	provided (Section 3.3)
2–3	Training / validation accuracy and loss curves	provided (Section 24)
4	Confusion matrix	provided for all six models (Section 25)
5, 10	Feature maps before / after attention	provided for A1–A5 (Section 28)
6	Channel-attention weights	provided (A1, A3, A5)
7–8	Spatial-attention heatmap and overlay	provided as overlay on the image (A2, A3, A5)
9	Self-attention response	provided (A4, A5)
11–12	Baseline and attention-model Grad-CAM	provided for all six models (Section 27)
13	Attention maps at P_2, P_3, P_4	**not provided** (P_3 only)
14	Graphical comparison of attention mechanisms	provided (Section 24)

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

Inference: Quantitative: accuracy 0.9220–0.9437, precision 0.9210–0.9483, recall 0.9223–0.9405 and macro F1 0.9214–0.9432; no model is significantly better than the baseline.

Table 17: Complexity of the six models relative to the baseline.

Model	Total params	Added params	Trainable (phase 1)	Trainable (fine-tune)	Infer. (ms/img)	Added time	Train time (min)
A0: Baseline ResNet50	23,597,957	–	10,245	14,963,717	7.10	–	14.1
A1: Channel Attention	23,730,117	+132,160 (0.56%)	142,405	15,095,877	7.68	+0.58 (8.2%)	15.3
A2: Spatial Attention	23,598,056	+99 (0.00%)	10,344	14,963,816	7.71	+0.61 (8.6%)	14.2
A3: Channel+Spatial	23,730,216	+132,259 (0.56%)	142,504	15,095,976	7.32	+0.22 (3.1%)	13.8
A4: Self-Attention	24,909,958	+1,312,001 (5.56%)	1,322,246	16,275,718	7.51	+0.41 (5.8%)	14.0
A5: Combined (C+S+SA)	25,042,217	+1,444,260 (6.12%)	1,454,505	16,407,977	7.96	+0.86 (12.1%)	15.9

Inference: Complexity: attention costs +99 to +1,444,260 parameters and +0.22 to +0.86 ms per image, but none of this buys a significant accuracy gain; A5 is the most expensive and the least accurate.

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

Accuracy + Precision + Recall + Macro F1

33.2 Complexity Analysis

Report:

Parameters + Inference Time

33.3 Qualitative Analysis

Report:

Grad-CAM + Attention Maps + Visual Interpretation

Therefore, the overall evaluation should be:

Quantitative + Computational + Qualitative

Inference: Qualitative: Grad-CAM and internal maps show saturated channel gating, border-dominated spatial gating and a single-hot-spot self-attention; the visible differences between models are image-dependent and agree with the small quantitative differences.

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.
5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

$$\boxed{\text{Attention Type}}$$

and

$$\boxed{\text{Attention Position}}.$$

Table 18: Status of the minimum mandatory ablation set.

No.	Required item (Section 35)	Status
1	Baseline ResNet50	done (A0)
2	Channel attention at P_3	done (A1)
3	Spatial attention at P_3	done (A2)
4	Channel+Spatial at P_3	done (A3)
5	Channel+Spatial at P_2	**not run**
6	Channel+Spatial at P_4	**not run**
7	Self-attention at P_3	done (A4)
8	Grad-CAM: baseline versus attention model	done (Section 27)
9	Spatial-attention visualisation at P_2, P_3, P_4	partly (P_3 only)
10	Graphical comparison of attention models	done (Section 24)

Inference: Items 5, 6 and 9 need the P_2 and P_4 models (B2, B4), which were not trained; the remaining items are covered by the six models.

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. What is the purpose of residual connections in ResNet50?
   Answer: They add the block input to its output, so each block learns only a residual and gradients flow directly through the shortcut, which makes very deep networks trainable.
2. What does conv2_x represent compared with conv5_x?
   Answer: conv2_x works on $56 \times 56 \times 256$ maps and captures edges, colours and textures; conv5_x works on $7 \times 7 \times 2048$ maps and encodes high-level, class-discriminative content.
3. Why does the spatial resolution decrease with network depth?
   Answer: Stride-2 convolutions (and the stem max-pool) downsample the maps to enlarge the receptive field and reduce computation.
4. Why does the number of feature channels increase with network depth?
   Answer: More filters are used in deeper stages to represent the growing variety of abstract patterns while the spatial size shrinks.
5. What is meant by low-level, intermediate, and high-level features?
   Answer: Low-level: edges, colours, textures; intermediate: parts such as petals and repeated structures; high-level: whole-object, class-specific information.

6. What is the purpose of an attention mechanism?
   Answer: To re-weight features so that informative channels, positions or relations are emphasised and less useful ones are suppressed at small extra cost.
7. What does channel attention attempt to learn?
   Answer: A per-channel importance vector ("what") computed from globally pooled channel statistics by a bottleneck MLP and a sigmoid.
8. What does spatial attention attempt to learn?
   Answer: A per-position weight map ("where") computed from channel-wise mean and max maps by a 7×7 convolution and a sigmoid.
9. How is channel attention different from spatial attention?
   Answer: Channel attention outputs one weight per channel shared over all positions; spatial attention outputs one weight per position shared over all channels.
10. Why can channel and spatial attention be used sequentially?
    Answer: They are complementary and shape-preserving, so channel attention can first select what matters and spatial attention then refine where it matters.
11. Why might channel attention behave differently at conv3_x and conv5_x?
    Answer: The channels of conv3_x (512, mid-level patterns) and conv5_x (2048, semantic patterns) represent different content, so the learned weights differ; this was not tested here (only P_3 was trained).
12. Why can spatial attention be useful at an intermediate ResNet stage?
    Answer: At P_2/P_3 the maps still have enough spatial resolution to localise flower parts while the features already carry meaningful structure; in our P_3 run the effect was small (A2 0.9437 versus 0.9347, not significant).
13. Why is self-attention computationally expensive when the feature map has high spatial resolution?
    Answer: It builds an $N \times N$ matrix with $N = HW$, so computation and memory grow quadratically with the number of positions.
14. What happens to the size of the self-attention matrix when the spatial resolution increases?
    Answer: It grows with the square of the number of positions: P_1 ($N = 3136$) 9,834,496 entries, P_2 ($N = 784$) 614,656, P_3 ($N = 196$) 38,416, P_4 ($N = 49$) 2,401.
15. How does attention performance vary when the insertion position is changed?
    Answer: Not evaluated: only P_3 was trained in this run.
16. How does attention at P_2 differ visually from attention at P_4?
    Answer: Not evaluated: only P_3 maps are available.
17. Does applying attention at multiple stages always improve classification performance?
    Answer: Not tested for several stages; stacking three modules at one stage (A5) gave the lowest accuracy (0.9220 versus 0.9347 for the baseline), so more attention is not automatically better.
18. Does Channel→Spatial attention behave differently from Spatial→Channel attention?
    Answer: Not evaluated: only Channel→Spatial (A3) was trained.
19. How does the spatial-attention kernel size influence the attention map?
    Answer: Not evaluated: only the 7×7 kernel was trained.
20. How does SE attention differ from ECA attention?
    Answer: SE uses two fully connected layers with a reduction ratio; ECA removes the bottleneck and uses a 1-D convolution over neighbouring channels with an adaptive kernel size, so it needs far fewer parameters (not run here).
21. How does CBAM combine channel and spatial attention?
    Answer: CBAM applies channel attention (a shared MLP on average- and max-pooled descriptors) and then spatial attention (a 7×7 convolution on channel-wise average and max maps), multiplying the features after each step.
22. How does self-attention differ from channel attention?
    Answer: Channel attention re-weights channels from global statistics without interaction between positions; self-attention computes pairwise relations between all positions and aggregates their values.
23. Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?
    Answer: No: A5 has the most parameters (+1,444,260) and the lowest accuracy (0.9220), while A2 adds 99 parameters and has the highest (0.9437); the differences are not significant.
24. What is the difference between an internal attention map and Grad-CAM?
    Answer: An internal attention map is a weight computed inside the forward pass and applied to the features; Grad-CAM is a gradient-weighted sum of feature maps that explains one class score after the prediction.
25. Why is Grad-CAM called a post-hoc explanation technique?
    Answer: It is computed after training and prediction from gradients, without changing the model or its computation.
26. Does the attention model suppress irrelevant background regions?
    Answer: Not clearly: the spatial gate is high over most of the image (low only near the borders), and Grad-CAM of the attention models still activates stems and grass in the tulip image.
27. Are the attention maps of correctly classified and incorrectly classified images different?

Answer: Grad-CAM of the misclassified images (Figure 10) still lies on the flower, so the maps differ mainly in the predicted class; internal attention maps of wrong predictions were not computed.

28. Which flower classes benefit most from attention?
    Answer: Roses and tulips, the weakest baseline classes (F1 gains up to +0.022 for roses and +0.039 for tulips); daisy lost F1 in all attention models.
29. What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?
    Answer: Which classes are confused and in which direction (e.g. baseline: 12 roses predicted as tulips, 4 tulips as roses), and per-class precision and recall, none of which overall accuracy shows.
30. Why is macro F1 useful in addition to accuracy?
    Answer: It averages the class-wise F1 with equal weight, so weak classes are not hidden by large ones (dandelion has 898 images, daisy 633).
31. What is meant by the accuracy–complexity trade-off?
    Answer: How much accuracy is gained per added parameter or computation; here +1.3–1.4 M parameters (A4, A5) brought no gain.
32. What is meant by the F1–inference-time trade-off?
    Answer: Whether additional latency brings better F1: A5 is slowest (7.96 ms) with the lowest F1, A2 gains 0.0084 F1 for +0.61 ms (not significant).
33. Why must every ablation study use the same dataset split?
    Answer: So that differences come from the model and not from different training or test images; with 551 test images, sampling alone changes accuracy by about ±2 points.
34. Why must the same training protocol be maintained when comparing different attention mechanisms?
    Answer: Otherwise optimiser, epochs, augmentation or seed would confound the effect of the attention type.
35. What conclusions can be drawn from the Grad-CAM comparison?
    Answer: Grad-CAM differences are image-dependent: A2 and A3 give more compact maps (0.549, 0.588 versus 0.496) without a significant accuracy gain, saturated predictions give blank maps, and errors still activate on the flower.

38. Observations to be Recorded

Students should record the following during experimentation:

1. Input and output feature dimensions for each attention module.
2. Training accuracy.
3. Validation accuracy.
4. Testing accuracy.
5. Training loss.
6. Validation loss.
7. Macro precision.
8. Macro recall.
9. Macro F1-score.
10. Number of trainable parameters.
11. Total number of model parameters.
12. Average inference time.
13. Confusion matrix.
14. Channel-attention weights.
15. Spatial-attention maps.
16. Self-attention responses.
17. Grad-CAM maps.
18. Attention maps for different insertion positions.

Table 19: Recorded observations (train/validation values of the best-validation checkpoint, evaluated without augmentation).

Model	In (P_3)	Out (P_3)	Train acc.	Val acc.	Test acc.	Train loss	Val loss	Prec.	Rec.	F1	Trainable	Total	ms/img
A0: Baseline ResNet50	(14,14,1024)	(14,14,1024)	0.9938	0.9418	0.9347	0.0190	0.2824	0.9385	0.9327	0.9347	14,963,717	23,597,957	7.10
A1: Channel Attention	(14,14,1024)	(14,14,1024)	0.9918	0.9400	0.9328	0.0248	0.3251	0.9316	0.9359	0.9329	15,095,877	23,730,117	7.68
A2: Spatial Attention	(14,14,1024)	(14,14,1024)	0.9961	0.9455	0.9437	0.0121	0.2707	0.9483	0.9405	0.9432	14,963,816	23,598,056	7.71
A3: Channel+Spatial	(14,14,1024)	(14,14,1024)	0.9879	0.9273	0.9383	0.0339	0.3238	0.9411	0.9379	0.9381	15,095,976	23,730,216	7.32
A4: Self-Attention	(14,14,1024)	(14,14,1024)	0.9942	0.9491	0.9328	0.0187	0.2965	0.9347	0.9314	0.9328	16,275,718	24,909,958	7.51
A5: Combined (C+S+SA)	(14,14,1024)	(14,14,1024)	0.9872	0.9418	0.9220	0.0366	0.3013	0.9210	0.9223	0.9214	16,407,977	25,042,217	7.96

Confusion matrices, channel-attention weights, spatial-attention maps, self-attention responses and Grad-CAM maps are given in Sections 25, 27 and 28; attention maps for other insertion positions were not recorded because those models were not run.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

40.1 Discussion of the Observed Results

- **Size of the effects.** The baseline already reaches 0.9347; the six models span 0.9220–0.9437, which is inside the ± 2 point sampling error of 551 test images, and no McNemar test against the baseline is significant (smallest $p = 0.349$).
- **Channel attention** (A1, A3, A5) produced nearly binary, nearly image-independent weights (same four top channels in all images), i.e. a static channel gate; it shifted errors between roses, tulips and dandelion/daisy rather than removing them.
- **Spatial attention** (A2) adds only 99 parameters; its map is high almost everywhere and low near the borders. It gave the highest accuracy and the most compact Grad-CAM, but the gain is not significant.
- **Self-attention** (A4) collapsed to one hot spot shared by all queries and a small γ; it converged fastest in the frozen phase because it adds 1,312,001 trainable parameters there, but ended at the baseline accuracy.
- **Combined attention** (A5) has the most parameters and the longest inference time and the lowest accuracy, so more attention is not better here.
- **Cost versus benefit.** None of the extra cost (up to +1.44 M parameters, +0.86 ms per image) is justified by a significant improvement on this dataset.
- **Quantitative versus qualitative.** More compact Grad-CAM maps (A2, A3) did not translate into significantly higher accuracy; saturated predictions give blank Grad-CAM maps, which limits the explanation of confident cases.
- **Limitations.** One seed and one run per model, only the P_3 position, no multi-stage/order/kernel experiments, no SE/ECA/CBAM comparison, Grad-CAM on softmax probabilities, and inference times from a shared Colab GPU.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.
Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.
A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

41.1 Conclusion

Six ResNet50 models (baseline, channel, spatial, channel+spatial, self-attention and combined attention at P_3) were trained and compared under an identical protocol on the flowers dataset. All reach 0.9220–0.9437 test accuracy; spatial attention is highest and combined attention lowest, but no difference to the baseline is statistically significant, and the larger modules (self-attention, combined) add parameters and latency without benefit. The effect of attention position, order, kernel size and the SE/ECA/CBAM mechanisms remains to be tested with the models marked "not run".

42. References

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